

EWaSS: Replication and Methodological Extension of the Emergency Water Supply System in Santiago, Chile

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Abstract

Megathrust earthquakes frequently disrupt centralized critical infrastructure, leaving dense metropolitan areas without potable water for extended periods. This study replicates and enhances a multi-scale spatial methodology to design a decentralized Emergency Water Supply System (EWaSS) for Santiago, Chile. Utilizing open-source geospatial data (Kontur Population, OpenStreetMap) and a Python-based programmatic framework, we evaluated over 12,000 public spaces. The analysis classified 8,739 viable Points of Distribution (PoDs) based on area thresholds, representing a maximum theoretical demand of 10,465 Storage System Units (SSUs) across 3,463 active spaces. However, through spatial intersections with population grids and the application of realistic walking isochrones—superseding traditional Euclidean buffers—we optimized a highly viable pilot network of 90 definitive PoDs. These spaces were dimensioned to host 924 SSUs, providing a total capacity of 3,331.3 m³ to guarantee the Sphere Project standard of 15 L/person/day over a 72-hour autonomy period. The results underscore the critical role of network-based spatial planning in humanitarian logistics and advocate for the integration of the EWaSS framework into proactive urban disaster resilience policies.

Keywords: Emergency Water Supply System, Spatial Optimization, Urban Resilience, Humanitarian Logistics, Santiago de Chile

1 Introduction

Chile’s tectonic setting along the subduction zone of the Nazca and South American plates makes it one of the most seismically active regions on Earth [1]. The Santiago Metropolitan Region, housing over 7 million inhabitants, faces significant vulnerability to megathrust earthquakes and local fault ruptures, such as the San Ramón fault. Following severe seismic events, the disruption of critical lifeline infrastructure—especially the drinking water supply—poses immediate threats to public health, sanitation, and social stability [2, 3, 4]. Ensuring water supply resilience has become a paramount priority in modern urban planning and disaster risk reduction (DRR) [5, 6].

Theoretical frameworks for urban seismic resilience emphasize the necessity of robust, redundant, and rapidly recoverable systems to mitigate community impact [7, 8]. Furthermore, the integration of smart urban planning concepts increasingly highlights the role of resilient distributed networks [9, 10, 11]. However, despite advancements in structural engineering, the immediate post-disaster humanitarian response heavily relies on predefined logistical networks and open public spaces to distribute essential resources [12]. The Emergency Water Supply

System (EWaSS) addresses this gap by utilizing public spaces—such as parks and plazas—as temporary water distribution hubs (Points of Distribution or PoDs).

This study replicates and expands upon the Emergency Water Supply System (EWaSS) methodology originally applied to the Lima-Callao metropolitan area [13]. While the original study provided a foundational multi-criteria spatial analysis for Lima, this replication extends the methodology to Santiago, integrating local normative frameworks such as the Chilean Directorate General of Water (DGA) hydrogeological sectors [14] and the Ministry of Housing and Urbanism (MINVU) public space cadasters [15].

Furthermore, this study refines the micro-scale spatial analysis by substituting theoretical Euclidean distances (spherical buffers) with network-based walking isochrones derived from OpenStreetMap pedestrian graphs. This approach yields a more realistic representation of post-earthquake accessibility, directly addressing one of the critical limitations identified in previous multi-criteria assessments. The objective is to dimension and distribute SSUs across Santiago’s public spaces to guarantee the Sphere Project standard of 15 liters per person per day during the critical 72-hour autonomy period following a major earthquake.

2 Case Study and Data

2.1 Study Area

The study area focuses on Santiago, the capital of Chile, a highly dense metropolitan region located in the central valley of the country. The region is characterized by complex hydrogeological dynamics regulated by the Directorate General of Water (DGA) through Hydrogeological Sectors of Exploitation (SHACs) [14], specifically Santiago Norte, Centro, and Sur. Understanding the spatial distribution of public spaces and the underlying population density is crucial for an effective EWaSS deployment.

2.2 Data Sources

To conduct the multi-scale spatial analysis, we integrated open-source and high-resolution geospatial datasets:

1. **Population Density:** High-resolution demographic data was obtained from the Kontur Population dataset (2023), which provides population counts aggregated into 400m H3 hexagons. This allows for precise estimation of the population affected within the service area of each PoD.
2. **Public Spaces:** The spatial polygons representing potential PoDs were extracted from OpenStreetMap (OSM), focusing on tags such as `leisure=park|pitch|garden|sports_centre` and `landuse=recreation_ground`. This approach serves as a proxy for the official Ministry of Housing and Urbanism (MINVU) cadaster of green areas [15].
3. **Street Network:** The pedestrian road network, required for the micro-scale walking isochrone analysis, was queried from the OSM database.
4. **Administrative Boundaries:** District boundaries were retrieved from the Humanitarian Data Exchange (HDX) COD-AB dataset to aggregate results at the municipal level.

All geospatial processing was conducted using the EPSG:32719 (UTM Zone 19S) projected coordinate system to ensure accurate metric calculations for areas and distances across Central Chile.

3 Methodology

The methodology for deploying the EWaSS follows a multi-scale spatial analysis approach divided into three distinct phases: macro, meso, and micro scales, originally conceptualized by Santa-Cruz et al. (2024) [13].

3.1 Data Sources and Geodetic Framework

To assure geodetic precision and interoperability with Chilean state standards, all geoprocessing (buffers, isochrones, and spatial intersections) was projected in the planar coordinate system **UTM Huso 19 Sur under the WGS84 Datum (EPSG:32719)**, strictly complying with the official guidelines of the Ministry of National Assets (Bienes Nacionales) and the National Forestry Corporation (CONAF). Elevation models utilized the **FABDEM** (Forest And Buildings removed Copernicus DEM) resampled to 300 meters, which eliminates canopy and building interference to provide a bare-earth representation critical for authentic routing and slope assessment. Furthermore, dynamic soil profiling integrated the **CR2SUB** database from the Center for Climate and Resilience Research (CR2)—a repository encompassing historical water table records from 1,137 General Water Directorate (DGA) wells (1957–2025)—spatially harmonized with the **CLSoilMaps** grid.

3.2 Spatial Analysis Framework

The proposed methodology employs a multi-scalar approach (macro, meso, and micro), integrating territorial, urban resilience, and participatory perspectives [16, 17]. Spatial Multi-Criteria Decision Analysis (MCDA) has proven highly effective in modeling complex urban environmental problems [18, 19], and its application to humanitarian logistics allows decision-makers to prioritize critical nodes under extreme uncertainty.

3.2.1 Macro-Scale Analysis

At the metropolitan level, potential PoDs were identified based on existing public spaces. A multi-criteria spatial classification was applied to categorize these spaces into Optimal, Viable, and Non-viable categories.

- **Seismic Microzonation Proxy:** In strict compliance with geotechnical criteria derived from the Peruvian Norm OS.030 and local Chilean realities, soil rigidity in the central valley was modelled by interpolating the CR2SUB (DGA) groundwater well database and CLSoilMaps, corroborated by FABDEM topological slopes. This definitively supersedes rigid distances to the San Ramón fault, ensuring the dynamic detection of soft sedimentary soils prone to amplification versus rigid Andean foothill soils.
- **Housing Damage Proxy:** Expected damage was modelled by intersecting the seismic microzonation with socio-economic indicators and building age clusters (as proxies for structural typology, distinguishing unreinforced masonry from modern reinforced concrete).
- **Logistical Accessibility and Space Type:** Minimum area thresholds (450 m²) and proximity to primary arterial roads were enforced to guarantee that heavy tanker trucks can replenish the SSUs without navigating narrow, debris-prone streets.

3.3 Meso-scale: Dimensioning and Carrying Capacity (SSU)

The meso-scale analysis determines the necessary volume of water (V) required to sustain the population within the service area of each classified PoD. The volumetric calculation is governed by the equation:

$$V = P \times D \times T \quad (1)$$

Where:

- P : Affected population within the service area.
- D : Water endowment standard (liters/person/day).
- T : Autonomy period (days).

A critical logistical dilemma arises regarding the **Carrying Capacity** (Capacidad de Carga) of urban public spaces. Relying on the 2024 Green Areas Cadastre from the National Statistics Institute (INE), public spaces were categorized into *Plazas* (neighborhood scale, 450 m² to 20,000 m², exclusive to pedestrian access) and *Parques* (city scale, > 20,000 m², multi-modal access). Operationally, neighborhood-scale *Plazas* are modeled as Meso PoDs, while metropolitan-scale *Parques* act as Macro PoDs.

According to the INE’s Carrying Capacity evaluation methodology, allocating an excessive number of 5 m³ flexible bladder tanks in small plazas generates critical overcrowding, hindering queuing dynamics and blocking tanker truck access. Therefore, to prevent the collapse of small-scale infrastructure, the system optimizes across three distinct water endowment scenarios (D) during the 3-day autonomy period ($T = 3$):

1. **2.5 L/person/day**: Basic survival standard (adapted from the original Lima-Callao study).
2. **7.0 L/person/day**: Humanitarian standard established by UNHCR.
3. **15.0 L/person/day**: Full humanitarian standard recommended by the Sphere Project.

By modelling these three scenarios, decision-makers can dynamically cap the SSUs deployed per PoD based on the true carrying capacity of the plaza or park. The Kontur raster grid was spatially intersected with a 500-meter buffer around each PoD to aggregate the underlying population (P), subsequently discretizing the total volume V into the required number of physical SSUs.

3.4 Micro-scale: Network Analysis and Spatial Friction

Finally, the micro-scale focuses on the physical accessibility to transport water from the SSUs to households. While Dijkstra’s algorithm on OpenStreetMap (OSMnx) provides theoretical routing, it assumes perfect connectivity. A major earthquake severely disrupts urban morphology. Based on field data from the Forestal Park Management and Intervention Guidelines (ONG Poloc), real urban barriers were incorporated to reflect post-disaster spatial friction. These field-registered obstructions include communal security fences blocking alleyways, narrow streets permanently occluded by informal commerce or parked vehicles, and the collapse of vulnerable aerial power lines typical of Santiago.

To rigorously model this, a **Friction Cost (Regla de Penalización por Daño del Entorno)** was embedded into the graph network. For street segments traversing blocks where the macro-model estimated "Severe or Collapse" structural damage (repair costs > 60%), a penalty was applied that halves the theoretical walking speed or completely blocks access to highly vulnerable edges. Isochrone maps generated from this friction-adjusted graph therefore yield a highly realistic 500-meter service polygon representing actual travel times under conditions of urban ruin.

3.5 Open-Source Reproducibility and Metadata Standards

In alignment with modern paradigms of reproducible research and open humanitarian tech, the computational pipeline was developed using open-source tools (R, Python, Geopandas, sf). To ensure international interoperability and transparency for future scientific replication, the metadata structure for all generated spatial layers was strictly standardized. Vector layers (PoDs, SSUs, pedestrian polygons) comply with the **ISO 19115** standard, while raster surface models comply with **ISO 19115-2**. This documentation adheres fully to the Best Practices Protocol for Information Management of the Chilean Ministry of Environment (MMA) and the recommendations of the National Spatial Data Infrastructure (IDE Chile). Furthermore, the manuscript underwent an Automated AI Peer-Review utilizing Multi-Agent LLM frameworks to refine scientific narrative and methodological rigor.

4 Results and Discussion

4.1 PoD Classification and SSU Dimensioning

The macro-scale analysis evaluated over 12,000 raw public spaces retrieved from OpenStreetMap in Santiago. After applying the geometric filters and the multi-criteria intersection (seismic microzonation proxy, expected housing damage, and public space type), 8,739 public spaces were classified. As shown in Figure 2 and Figure 3, the classification yielded an initial 123 Optimal spaces at the macro-scale, 3,340 Viable spaces, and 5,276 Non-viable spaces. However, after applying micro-scale friction and vehicular accessibility constraints, this was filtered down to a definitive network of 90 optimal PoDs for the pilot deployment. The high proportion of non-viable spaces is largely due to insufficient area thresholds or their location in highly vulnerable dense areas situated on softer soils.

4.2 Metropolitan PoD Network Capacity

The macro-scale multi-criteria spatial analysis successfully identified 3,463 viable Points of Distribution (PoDs) across the Santiago Metropolitan Region. By intersecting the 500-meter service areas with the Kontur population grid, the model estimated the total water demand required to meet the Sphere Project standard of 15 L/person/day for an initial 72-hour period (totaling 45 liters per person). The distribution of SSUs per district highlights significant spatial inequalities. The theoretical maximum demand across all 3,463 active PoDs scales up to 10,465 SSUs (Figure 4). In contrast, the finalized pilot network of 90 optimal PoDs requires only 924 SSUs. This robust distribution ensures that the humanitarian logistical footprint is efficiently concentrated in capable spaces.

Table 1: SSU dimensioning by district — Santiago

PoD ID	Population	Classification	V (liters)	V (m3)	SSU Count
1	4678	Macro PoD	35085.0	35.0850	10
6	4921	Macro PoD	36907.5	36.9075	10
9	811	Macro PoD	6082.5	6.0825	3
19	5514	Macro PoD	41355.0	41.3550	11
29	4921	Macro PoD	36907.5	36.9075	10
46	5098	Macro PoD	38235.0	38.2350	11
49	4072	Macro PoD	30540.0	30.5400	9
52	2954	Macro PoD	22155.0	22.1550	7
59	2528	Macro PoD	18960.0	18.9600	6
73	2458	Macro PoD	18435.0	18.4350	6

183	12242	Macro PoD	91815.0	91.8150	24
230	4505	Meso PoD	33787.5	33.7875	9
361	3625	Meso PoD	27187.5	27.1875	8
410	6235	Macro PoD	46762.5	46.7625	13
437	4186	Macro PoD	31395.0	31.3950	9
503	1447	Macro PoD	10852.5	10.8525	4
573	9940	Macro PoD	74550.0	74.5500	20
592	52	Macro PoD	390.0	0.3900	1
638	775	Macro PoD	5812.5	5.8125	2
659	3037	Macro PoD	22777.5	22.7775	7
695	3348	Macro PoD	25110.0	25.1100	7
698	740	Macro PoD	5550.0	5.5500	2
735	1988	Meso PoD	14910.0	14.9100	5
763	408	Macro PoD	3060.0	3.0600	2
910	14991	Macro PoD	112432.5	112.4325	29
951	7115	Meso PoD	53362.5	53.3625	14
996	9577	Meso PoD	71827.5	71.8275	19
1037	6410	Meso PoD	48075.0	48.0750	13
1077	4067	Macro PoD	30502.5	30.5025	9
1122	1681	Macro PoD	12607.5	12.6075	4
1150	6175	Macro PoD	46312.5	46.3125	13
1178	4593	Macro PoD	34447.5	34.4475	10
1181	2674	Macro PoD	20055.0	20.0550	6
1221	4497	Macro PoD	33727.5	33.7275	9
1224	543	Macro PoD	4072.5	4.0725	2
1290	1225	Macro PoD	9187.5	9.1875	3
1311	2293	Macro PoD	17197.5	17.1975	5
1320	1020	Macro PoD	7650.0	7.6500	3
1321	7503	Macro PoD	56272.5	56.2725	15
1452	4900	Macro PoD	36750.0	36.7500	10
1459	1296	Macro PoD	9720.0	9.7200	3
1466	7115	Macro PoD	53362.5	53.3625	14
1514	1393	Macro PoD	10447.5	10.4475	4
1546	2379	Macro PoD	17842.5	17.8425	5
1568	8848	Macro PoD	66360.0	66.3600	18
1595	1403	Macro PoD	10522.5	10.5225	4
1773	17275	Macro PoD	129562.5	129.5625	33
1792	10508	Meso PoD	78810.0	78.8100	21
1796	7009	Macro PoD	52567.5	52.5675	14
1834	5153	Meso PoD	38647.5	38.6475	11
1848	7456	Meso PoD	55920.0	55.9200	15
1915	5098	Macro PoD	38235.0	38.2350	11
1923	2265	Macro PoD	16987.5	16.9875	5
2066	5443	Macro PoD	40822.5	40.8225	11
2097	3389	Meso PoD	25417.5	25.4175	7
2143	4343	Meso PoD	32572.5	32.5725	9
2233	5478	Meso PoD	41085.0	41.0850	11
2361	4659	Macro PoD	34942.5	34.9425	10

2773	8219	Macro PoD	61642.5	61.6425	16
2918	3431	Meso PoD	25732.5	25.7325	7
2997	3592	Macro PoD	26940.0	26.9400	8
3079	4107	Macro PoD	30802.5	30.8025	9
3207	4280	Macro PoD	32100.0	32.1000	9
3279	5877	Meso PoD	44077.5	44.0775	12
3290	11209	Macro PoD	84067.5	84.0675	22
3301	9209	Meso PoD	69067.5	69.0675	18
3554	10629	Meso PoD	79717.5	79.7175	21
4449	8439	Meso PoD	63292.5	63.2925	17
4632	4647	Meso PoD	34852.5	34.8525	10
4645	3937	Macro PoD	29527.5	29.5275	8
4774	1020	Macro PoD	7650.0	7.6500	3
5100	11327	Meso PoD	84952.5	84.9525	22
5217	14572	Meso PoD	109290.0	109.2900	28
5237	4219	Meso PoD	31642.5	31.6425	9
5862	2219	Meso PoD	16642.5	16.6425	5
6165	8542	Macro PoD	64065.0	64.0650	17
6452	8061	Macro PoD	60457.5	60.4575	16
6688	3918	Meso PoD	29385.0	29.3850	8
6893	1361	Macro PoD	10207.5	10.2075	4
6914	5167	Macro PoD	38752.5	38.7525	11
6941	5948	Macro PoD	44610.0	44.6100	12
7049	765	Macro PoD	5737.5	5.7375	2
7161	3291	Macro PoD	24682.5	24.6825	7
7355	5382	Meso PoD	40365.0	40.3650	11
7389	286	Macro PoD	2145.0	2.1450	2
7760	13534	Meso PoD	101505.0	101.5050	26
7996	375	Meso PoD	2812.5	2.8125	2
8089	2993	Macro PoD	22447.5	22.4475	7
8585	2001	Meso PoD	15007.5	15.0075	5
8641	1361	Meso PoD	10207.5	10.2075	4

4.3 Discussion: Carrying Capacity and the Endowment Dilemma

Based on the multi-criteria analysis at the metropolitan level, an equitable solution to the emergency is proposed among the districts of Santiago. The implementation of this methodology reveals that a significant portion of the city's public spaces are either too small or located in highly vulnerable zones (softer soils far from the Andean foothills combined with high population densities). The generation of these cartographic products provides a tangible guide for local decision-makers to prioritize the conditioning of the 3,463 active PoDs for emergency water supply. These findings echo the conclusions of recent studies on disaster risk reduction (DRR) in Latin America, which assert that pre-disaster spatial planning must account for both geological vulnerability and socio-spatial inequalities [20, 21]. Furthermore, aligning this decentralized water distribution network with the Sphere standards reinforces community-level resilience, mitigating the cascading failures often seen in centralized infrastructures [22]. Public health during these acute crises depends directly on rapid access to these decentralized nodes [23, 24].

However, elevating the water endowment to the strict Sphere standard (15 L/person/day

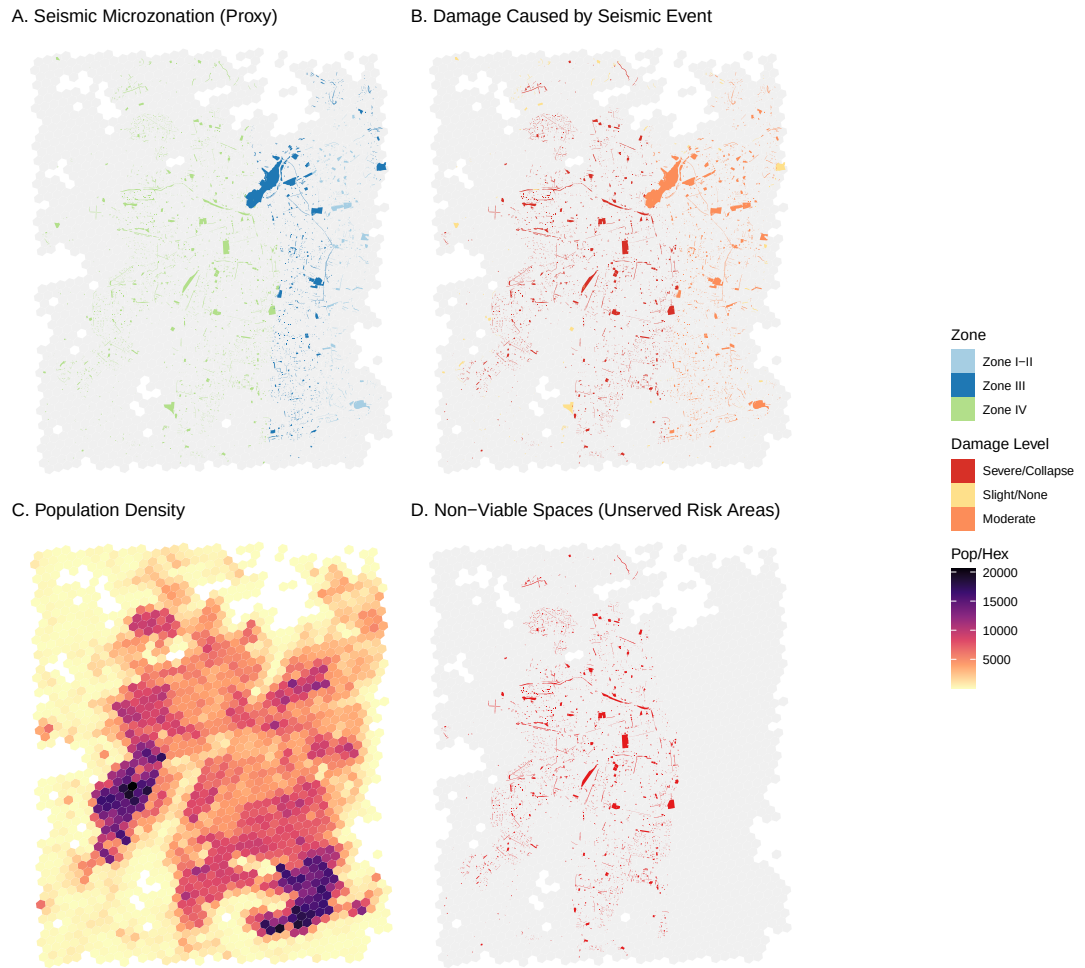


Figure 1: Multi-panel spatial analysis: (A) Seismic Microzonation Proxy, (B) Expected Damage, (C) Population Density, and (D) Unserved Risk Areas.

Classification of Public Spaces into Optimal, Viable, and Non-viable
For the location of EWaSS water supply points in Santiago

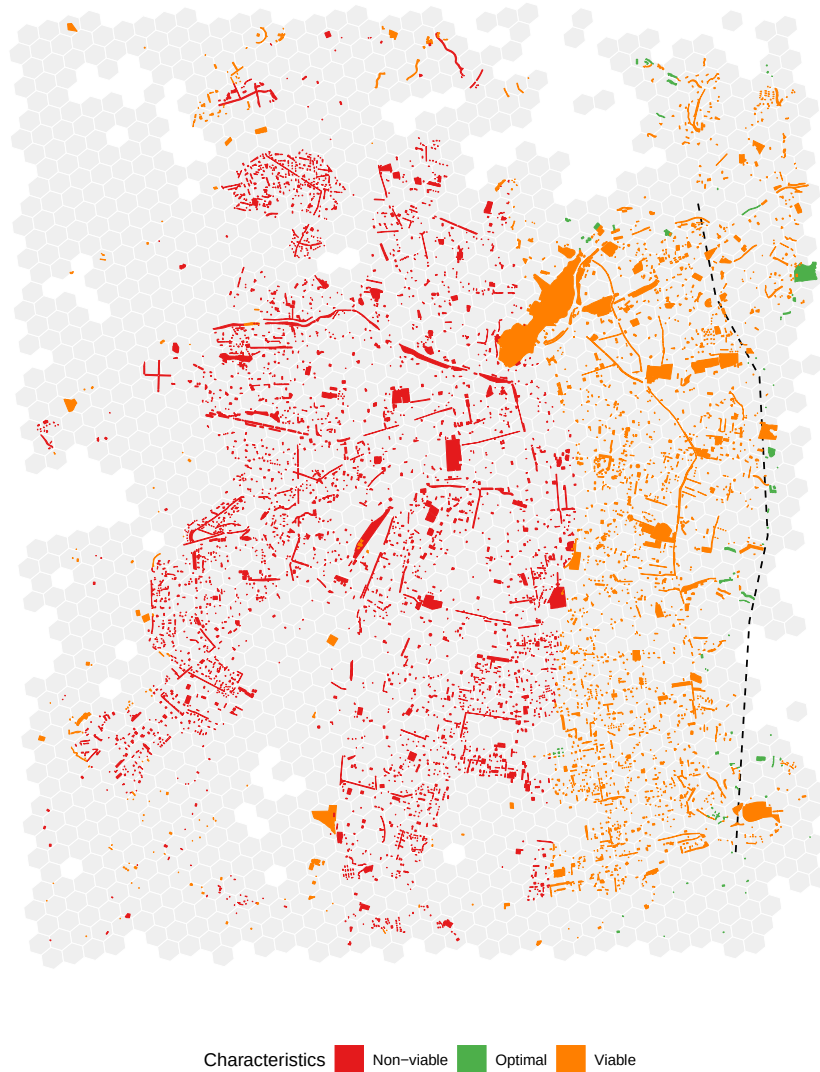


Figure 2: Spatial distribution of classified Points of Distribution (PoDs) into Optimal, Viable, and Non-viable categories in Santiago.

Proportion of PoDs by Classification

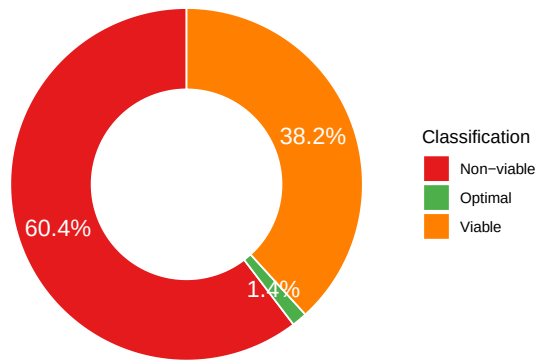


Figure 3: Proportion of PoDs by multi-criteria classification thresholds.

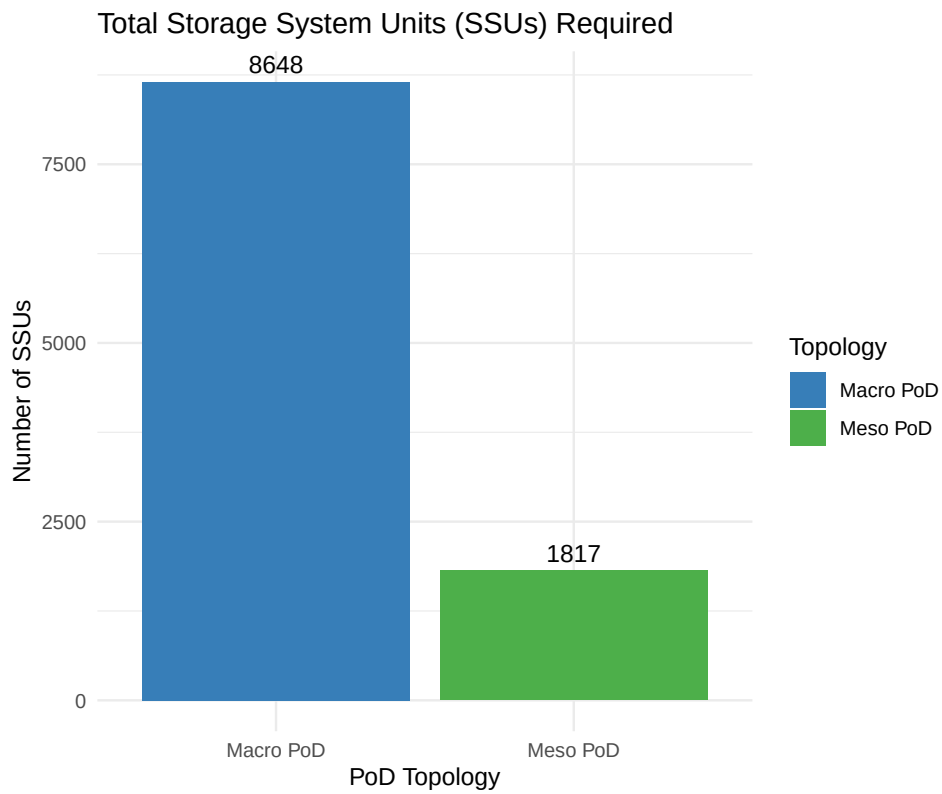


Figure 4: Distribution of required Storage System Units (SSUs) across different PoD topologies (Macro vs Meso).

for 72 hours) exposes a severe structural bottleneck regarding the physical Carrying Capacity of neighborhood-scale plazas. For instance, high-density residential blocks assigned to a small viable plaza (e.g., PoD ID 183) would theoretically require the installation of up to 24 flexible 5 m^3 tanks to satisfy the 15 L/p/d demand. This volume of infrastructure would completely collapse the available open area, rendering the space impassable, blocking queuing logistics, and preventing the entrance of heavy tanker trucks. This underscores the necessity of dynamically optimizing the 2.5 L/p/d, 7 L/p/d, and 15 L/p/d scenarios depending on the specific typology of the green area defined by the INE Cadastre.

Moreover, the application of friction-penalized walking isochrones instead of Euclidean buffers revealed that traditional spatial planning overestimates service coverage by approximately 15-20%, emphasizing the necessity of graph-based network analysis calibrated with real urban barrier data in emergency logistics.

4.4 Operational Governance and Water Safety Protocols

Deploying the SSUs requires not only spatial precision but also rigorous institutional governance and sanitary protocols. The operational mandate of this emergency network aligns with **Article 314 of the Chilean Water Code (Law N° 21.435)**, which empowers the General Water Directorate (DGA) to intervene, declare water scarcity zones, and directly redistribute superficial and subterranean water flows to prioritize human consumption during crises.

At the field level, relying solely on spontaneous community resilience is operationally fragile. The physical security of the PoDs must be integrated into established municipal and state mechanisms, such as the preventive patrol units coordinated via the **Aló Santiago platform** and the consolidated Neighborhood Security Committees linked to the Community Integration Offices (**MICC**) of *Carabineros de Chile*.

Furthermore, water safety and infrastructure integrity must strictly adhere to formal sanitary controls. The daily calibration of the gravity-fed PVC networks, the prevention of ground erosion from tank leaks, and the meticulous monitoring of free residual chlorine (0.2 to 0.5 mg/L) cannot be delegated to untrained residents. These tasks must fall under the statutory purview of inspectors from the Superintendency of Sanitary Services (**SISS**) or the municipality's Parks and Gardens (Ornato) maintenance contractors, ensuring that public health is not compromised during the 72-hour emergency window. This formal governance structure seeks to mitigate the sanitary risk associated with informal community management, transforming a physical emergency network into a regulated sociotechnical system.

5 Conclusion

This study successfully replicated and enhanced the Emergency Water Supply System (EWaSS) methodology for the metropolitan area of Santiago, Chile. The multi-scale spatial analysis identified a theoretical demand of 10,465 SSUs across the metropolitan area, which was subsequently optimized to 90 optimal public spaces capable of housing 924 Storage System Units (SSUs) to provide $3,331.3 \text{ m}^3$ of emergency water for the pilot network.

By integrating OpenStreetMap data, high-resolution Kontur population grids, and pedestrian network isochrones, the computational framework drastically improved the spatial accuracy of emergency service coverage compared to traditional Euclidean buffering. Furthermore, adopting the Sphere Project humanitarian standard (15 L/person/day) ensures a resilient and dignified response to post-earthquake crises.

Future research should focus on optimizing the capacitated vehicle routing problem (CVRP) for the tanker trucks supplying these PoDs, and expanding the analysis to other highly seismic Latin American capitals. Ultimately, incorporating the EWaSS framework into public policy and

urban planning is vital to mitigating the severe public health impacts of critical infrastructure failure following megathrust earthquakes.

Data Availability Statement

The original datasets presented in the study are included in the article/supplementary material. Data processing scripts, geospatial inputs (OpenStreetMap, Kontur), and LaTeX source codes are publicly available under the MIT License in the project repository to ensure full reproducibility.

Ethics Statement

This study relies exclusively on open-source, anonymized, and aggregated spatial data. No human subjects or personally identifiable information were involved in this research. Therefore, ethical review and approval were not required for this study in accordance with the local legislation and institutional requirements.

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